FIELD-READY TRAINING SYSTEM
DRONE
ZERO TO HERO
A practical UAV operator crash course
PRESENTATION / FIELD ACTIVITIES / ASSESSMENTS
10 40 05
MODULES HOURS DAYS
LEARNER EDITION
Designed for competency-based introductory RPAS training

COURSE GUIDE
The operating philosophy
SKILL FIRST. THEORY ONLY WHEN IT CHANGES A DECISION.
Teach like an intensive practical discipline: show one action, rehearse it safely, challenge it under controlled conditions, assess it, then advance. Never add complexity before the previous safety gate is passed.
DEMONSTRATE / PRACTICE / CHALLENGE / EVALUATE
What trainees can do at completion
Make a documented GO / NO-GO decision and conduct a crew brief.
Inspect, prepare, take off, maneuver, land and log a training drone.
Recognize common abnormal states and choose safe recovery.
Collect usable aerial imagery and verify a basic mission/map output.
Evaluate current AI, RTK and sensing tools without overstating their capabilities.
Important regulatory boundary
An institutional course completion record is NOT a CAAP-issued Controller Certificate, RPAS Operator Certificate, permission to conduct commercial operations, or authorization for BVLOS/night/restricted flights.
Training organizations must check applicable CAAP provider requirements, including MC No. 064-2026, befo delivery.
Publication date: 30 October 2026 | Local context: Philippines | Review: before each cohort and when rules/software change.

COURSE GUIDE
Five-day pathway
DAY 01 Safety, aircraft, first flight
M1 2h + M2 3h + M3 3h
DAY 02 First flight and precision
M3 2h + M4 6h
DAY 03 Precision, emergency, camera
M4 1h + M5 5h + M6 2h
DAY 04 Camera, autonomy and AI
M6 2h + M7 5h + M8 1h
DAY 05 Future systems, mission, checkride
M8 2h + M9 4h + M10 2h
Delivery and training time
Target blend: 75% applied activity, simulator, briefing, field work and data review; 25% essential teaching. Aim for at
least 5 hours of individually logged supervised live-aircraft manipulation per trainee, separately from simulator time.
Actual completion depends on weather, site permission, instructor ratio and fleet availability.

COURSE GUIDE
Assessment and release-to-flight

SAFETY GATE
Critical hazards or checklist failures stop assessment, regardless of points.
AIRCRAFT GATE
Correct controls, battery, propellers, firmware/alerts and return settings.
CONTROL GATE
Smooth stable hover, translation, orientation, safe landing and communications.
MISSION GATE
Approved plan, safe supervision, usable data and limitations statement.
Recommended overall weighting
Safety & aircraft preparation 25%
Manual flight proficiency 30%
Emergency judgment 20%
Mission execution and data quality 25%
Pass: >=80% total plus PASS in every critical item. Reassess after targeted practice. The performance rubrics are instructional
and do not substitute for official CAAP assessment rules.

MODULE 01
LEARNING MODULE
DRONE SURVIVAL RULES
Safety before flight
END-OF-MODULE COMPETENCY
Make a defensible GO / NO-GO decision for a proposed flight.
2 HOURS CLASSROOM + FIELD
Performance targets
Check operational purpose, aircraft weight and pilot/provider obligations.
Identify airspace, people, structures, weather and manned-aircraft exposure.
Brief pilot, observer, boundaries and emergency landing points.
Presentation sequence
01 02 03
Every launch is a decision What CAAP expects Airspace and other aircraft
04 05 06
Weather is a flight control Build your safe flight box The one-minute brief
ADVANCE ONLY WHEN
No unaddressed critical hazard; correct decision; clear roles and stop triggers.

MODULE 01
PRESENTATION
Module 01 / Slides 1-3

Safety before flight
SLIDE 01 Every launch is a decision

A safe flight begins before motors start.
Define the mission and its expected benefit.
Check authority, aircraft, crew and site.
Separate essential risk from optional convenience.
DO: Ask: What could force a NO-GO right now?
WATCH: A drone app saying Ready does not mean legal or safe.

SLIDE 02 What CAAP expects

Do not guess regulatory permission.
Classify the operation: commercial or non-commercial.
Check CAAP rules, local restrictions and required certificates/permits.
Verify approvals for special operations before arriving.
DO: Demonstrate finding CAAP Types of Operations and RPAS Regulations pages.
WATCH: Recheck requirements for each site and mission.

SLIDE 03 Airspace and other aircraft

Manned aircraft always take priority.
Identify airports, helipads, approach routes and temporary restrictions.
Watch for low-flying emergency, military or agricultural aircraft.

Yield, descend or land safely when conflict develops.
DO: Mark hazards on a printed field map.
WATCH: Map apps can lag changes in airspace restrictions.

MODULE 01
PRESENTATION
Module 01 / Slides 4-6

Safety before flight
SLIDE 04 Weather is a flight control

Wind and visibility change the risk envelope.
Check surface wind, gusts, rain risk and visibility.
Consider sun glare, terrain turbulence and changing conditions.
Apply aircraft-specific operating limits with margin.
DO: Use a GO / NO-GO board for two weather scenarios.
WATCH: Never teach flying into unsafe winds for confidence.

SLIDE 05 Build your safe flight box

People, obstacles and separation matter.
Define takeoff/landing area, flight limits and spectator zone.
Assign pilot-in-command and observer duties.
Choose emergency landing options and abort triggers.
DO: Walk site perimeter and identify sterile zones.
WATCH: Avoid flight over uninvolved people.

SLIDE 06 The one-minute brief

Communication prevents avoidable incidents.
Mission and time limit; flight area and altitude plan.
Who calls traffic, batteries, hazards and emergency landing.
Clear verbal commands: HOLD, ABORT, LAND, CLEAR.
DO: Role-play the briefing with pilot and spotter.
WATCH: Everyone should know who can call STOP.

MODULE 01
PRACTICAL ACTIVITY
Site GO / NO-GO reconnaissance
Module 01 | Est. 35 min

TASK / DEMONSTRATION
Make a defensible GO / NO-GO decision for a proposed flight.
Equipment and setup
Field map, weather app, CAAP guidance, cones and risk card.
Procedure
Receive a site scenario with one hidden hazard.
1
Mark boundary, obstacles, people, airspace questions and emergency landing options.
2
Write GO, CONDITIONAL GO, or NO-GO with three reasons.
3
Deliver a 60-second crew brief and defend the decision.
4
EVIDENCE TO SUBMIT
Completed risk card, annotated site map and signed go/no-go record.
PASS CONDITION
No unaddressed critical hazard; correct decision; clear roles and stop triggers.
DEBRIEF: Which single new fact would reverse your decision?
SAFETY LIMIT: Do not treat a classroom exercise as actual operational clearance.

MODULE 01
ASSESSMENT

Module 01 / Knowledge check

Circle one best answer. 5 questions x 1 point. No guessing in flight-critical tasks.
Name _____ Date _____
1 A map shows a nearby airfield but the drone displays Ready. What next?
A. Launch at low altitude B. Verify airspace restrictions and permissions
C. Fly only for five minutes D. Ask a spectator
2 When manned aircraft approaches the operating area, the pilot should:
A. Hold altitude and continue B. Climb above it
C. Yield and take safe avoidance/landing action D. Use auto-follow
3 Who should be authorized to call STOP during the brief?
A. Only the photographer B. Any designated crew member identifying danger
C. Only the client D. Only the trainee
4 A strong gust exceeds the aircraft limit. The correct decision is:
A. Use sport mode B. Reduce camera resolution
C. Postpone or cancel the flight D. Fly behind a tree
5 A GO / NO-GO review is complete when:
A. The app connects B. Battery reaches 100%
C. Risks, authority, site and crew are checked D. The weather is sunny
Written questions check understanding; the activity performance gate determines whether the trainee may progress. Answer
key and rationale are placed in the instructor section.

MODULE 02
LEARNING MODULE
KNOW YOUR MACHINE
Setup and inspection
END-OF-MODULE COMPETENCY
Prepare, inspect and safely configure a training drone without prompting.
3 HOURS BENCH + FIELD
Performance targets
Recognize critical aircraft, power, navigation and communication components.
Perform an aircraft-specific inspection and battery check.
Verify home point, flight modes, geofence and loss-link behavior.
Presentation sequence
01 02 03
The flight system Power is an operating limit Inspect before arming
04 05 06
The home point trap Know the flight mode Ready means cross-checked
ADVANCE ONLY WHEN
Detect every critical simulated defect and complete the approved checklist in order.

MODULE 02
PRESENTATION
Module 02 / Slides 1-3

Setup and inspection
SLIDE 01 The flight system

Know what each part is responsible for.
Airframe, motors, props and power deliver thrust.
Flight controller and IMU stabilize attitude.
GNSS, compass, vision sensors and radios support navigation.
DO: Point to each component on the actual aircraft.
WATCH: A GNSS receiver does not guarantee perfect position.

SLIDE 02 Power is an operating limit

Batteries require discipline, not guesswork.
Inspect housing, swelling, contacts and temperature.
Use the correct charger and manufacturer procedures.
Plan landing reserve from conditions and mission length.
DO: Show examples of safe/unsafe battery findings.
WATCH: Do not use damaged or swollen packs.

SLIDE 03 Inspect before arming

A repeatable checklist beats memory.
Check propellers, mounts, arms and payload security.
Confirm controller link, firmware compatibility and warnings.
Verify takeoff site and nearby people are clear.
DO: Instructor demonstrates a deliberate missing-propeller defect.
WATCH: Never spin unguarded propellers on a workbench.

MODULE 02
PRESENTATION
Module 02 / Slides 4-6

Setup and inspection
SLIDE 04 The home point trap

Return-to-Home only helps when configured well.
Verify recorded home point and position quality.
Set safe return altitude for obstacles and terrain.
Know what happens on low battery, lost link and manual cancel.
DO: Compare poor and good RTH plans on a site sketch.
WATCH: RTH is not collision-proof or universally available.

SLIDE 05 Know the flight mode

Automation changes pilot workload.
Distinguish position-hold, altitude-hold and manual-assist behavior.
Know which modes depend on GNSS/vision sensors.
Practice identifying current mode, battery and signal.
DO: Use the controller screen to read three status fields.
WATCH: Switching modes without understanding behavior increases risk.

SLIDE 06 Ready means cross-checked

The pilot accepts the final decision.
Read manufacturer warnings and respond to each.
Confirm observer, crew brief, battery, area and mode.
Use a spoken READY / CLEAR / TAKEOFF call.
DO: Run the complete checklist with a second trainee watching.
WATCH: Do not dismiss unexplained calibration or sensor errors.

MODULE 02
PRACTICAL ACTIVITY
Aircraft turnaround challenge
Module 02 | Est. 55 min

TASK / DEMONSTRATION
Prepare, inspect and safely configure a training drone without prompting.
Equipment and setup
Training drone, controller, battery, propeller set, checklist, defect cards.
Procedure
Identify ten major aircraft and controller parts.
1
Inspect a grounded aircraft containing instructor-inserted nonhazardous defect cards.
2
Demonstrate safe battery installation and status verification.
3
Record home-point, RTH/flight limits and failsafe checks.
4
Brief release-to-flight or reject with reasons.
5
EVIDENCE TO SUBMIT
Signed turnaround sheet with defects and corrective actions.
PASS CONDITION
Detect every critical simulated defect and complete the approved checklist in order.
DEBRIEF: What is the most dangerous warning to ignore?
SAFETY LIMIT: All faults are staged as tags or non-powered mockups; no unsafe flight configurations.

MODULE 02
ASSESSMENT

Module 02 / Knowledge check

Circle one best answer. 5 questions x 1 point. No guessing in flight-critical tasks.
Name _____ Date _____
1 The principal job of the flight controller is to:
A. Store videos B. Control stabilization and flight behavior
C. Charge batteries D. Replace visual observation
2 A battery has visible swelling. Best action?
A. Fly a short sortie B. Discharge it in flight
C. Remove from service and follow battery safety procedure D. Freeze it
3 Why confirm the home point?
A. It improves camera color B. Autonomous return may use it
C. It boosts radio power D. It increases battery capacity
4 Which is a valid pre-arm observation?
A. All propellers are undamaged and correctly installed B. A missing propeller is harmless
C. Spectators can stand anywhere D. Ignore all alerts
5 A mode relying on GNSS should be used when:
A. The display says any mode is okay B. Required positioning sensors are valid
C. The drone is indoors with no navigation support D. The propellers are removed
Written questions check understanding; the activity performance gate determines whether the trainee may progress. Answer
key and rationale are placed in the instructor section.

MODULE 03
LEARNING MODULE
FIRST FLIGHT: MASTER THE CONTROLS
Launch, hover, land
END-OF-MODULE COMPETENCY
Safely take off, hold position, translate and land under supervision.
5 HOURS SIMULATOR + LIVE
Performance targets
Identify yaw, throttle, pitch and roll on the configured control scheme.
Control hover and basic translation using smooth inputs.
Land safely and disarm using the aircraft procedure.
Presentation sequence
01 02 03
The four control axes Eyes outside, data inside Takeoff sequence
04 05 06
Hover is a skill Translate, stop, repeat Land and secure
ADVANCE ONLY WHEN
No safety intervention; stable 30-second hover; controlled landing; correct commands.

MODULE 03
PRESENTATION
Module 03 / Slides 1-3

Launch, hover, land
SLIDE 01 The four control axes

Understand movement before moving sticks.
Throttle changes lift/height depending on flight mode.
Yaw rotates heading; pitch moves fore/aft; roll moves sideways.
Practice with the exact transmitter mode used in training.
DO: Map stick inputs on a simulator with props removed from live aircraft.
WATCH: Do not assume stick layouts are identical across equipment.

SLIDE 02 Eyes outside, data inside

Scan the aircraft and reference points.
Keep visual orientation and separation.
Use telemetry for battery, mode, altitude and warnings.
Take short, deliberate looks at the screen.
DO: Alternate 5-second external scan and status confirmation.
WATCH: Do not fly by the screen alone for VLOS training.

SLIDE 03 Takeoff sequence

Small, confident and controlled.
Confirm CLEAR command and stable system status.
Lift steadily to a safe supervised training hover.
Hold position and assess drift, vibration and control response.
DO: Instructor shows one correct and one aborted simulated launch.
WATCH: Never take off when the area is not clear.

MODULE 03
PRESENTATION
Module 03 / Slides 4-6

Launch, hover, land
SLIDE 04 Hover is a skill

Stability needs constant awareness.
Use small inputs; neutralize to stop drift when appropriate.
Keep orientation known and a safe horizontal margin.
Respond to wind within approved operating limits.
DO: Use a ground reference marker without flying directly over observers.
WATCH: Do not chase errors with large, abrupt stick movements.

SLIDE 05 Translate, stop, repeat

Build dependable movement patterns.
Move forward and backward in a defined flight box.
Move left and right while maintaining heading.
Return to center and stop with smooth control.
DO: Trainees call intention before each movement.
WATCH: Maintain separation from other operating aircraft.

SLIDE 06 Land and secure

The sortie ends after post-flight checks.
Confirm landing site and approach path.
Descend steadily and verify touchdown/disarm.
Check battery, airframe and log anomalies.
DO: Practice positive verbal landing callouts.
WATCH: Never approach moving propellers.

MODULE 03
PRACTICAL ACTIVITY
Takeoff - hover - box - land
Module 03 | Est. 120 min including rotation

TASK / DEMONSTRATION
Safely take off, hold position, translate and land under supervision.
Equipment and setup
Simulator, approved training drone, marked open field, spotter, log sheet.
Procedure
Pass control-axis familiarization in simulator.
1
Complete preflight and call CLEAR / TAKEOFF.
2
Maintain supervised hover for 30 seconds.
3
Fly 4 controlled translations, returning to center after each.
4
Land inside the designated safe zone and complete postflight.
5
EVIDENCE TO SUBMIT
Instructor flight card: simulator result, hover time and individual live-aircraft log.
PASS CONDITION
No safety intervention; stable 30-second hover; controlled landing; correct commands.
DEBRIEF: Which control direction caused your largest correction?
SAFETY LIMIT: Instructors select heights, weather limits and separation based on aircraft, location and trainee ability.

MODULE 03
ASSESSMENT

Module 03 / Knowledge check

Circle one best answer. 5 questions x 1 point. No guessing in flight-critical tasks.
Name _____ Date _____
1 Yaw changes the aircraft's:
A. Heading B. Battery temperature
C. Camera resolution D. GPS coordinates directly
2 A good correction to mild drift is usually:
A. A large stick snap B. A small measured control input
C. Immediate sport mode D. Stop observing
3 What should be confirmed before live takeoff?
A. Video quality only B. Clear area, approved flight mode and readiness
C. High-speed mode D. Maximum height
4 During VLOS training the trainee should:
A. Watch only the screen B. Keep the aircraft visually observed
C. Ignore observer calls D. Follow the aircraft on foot
5 After touchdown, the next action is:
A. Reach toward spinning propellers B. Verify disarm and secure aircraft
C. Remove battery while motors run D. Immediately launch again
Written questions check understanding; the activity performance gate determines whether the trainee may progress. Answer
key and rationale are placed in the instructor section.

MODULE 04
LEARNING MODULE
PRECISION FLIGHT
Orientation and maneuvers
END-OF-MODULE COMPETENCY
Fly defined patterns and manage nose-in orientation with precision.
7 HOURS SIMULATOR + LIVE
Performance targets
Use controlled stick inputs through changing orientation.
Fly square, figure-eight and precision-landing patterns.
Respect defined boundaries and demonstrated wind limits.
Presentation sequence
01 02 03
Orientation changes everything Build the square Figure-eight coordination
04 05 06
Nose-in confidence Precision landing Flight discipline
ADVANCE ONLY WHEN
Route remains inside approved flight box; no critical correction; controlled landing within 1 m when safe.

MODULE 04
PRESENTATION
Module 04 / Slides 1-3

Orientation and maneuvers
SLIDE 01 Orientation changes everything

Aircraft-relative commands can feel reversed.
Nose-out feels intuitive; nose-in often reverses lateral perception.
Use tiny inputs and keep a visual reference.
Train nose-in in simulator before the live-aircraft gate.
DO: Show two identical roll inputs at opposite headings.
WATCH: Panic corrections are often larger than the original drift.

SLIDE 02 Build the square

Edges teach planning and discipline.
Select safe corner references in an open area.
Fly straight legs, pause and rotate deliberately.
Prioritize control over speed.
DO: Trace route on the field diagram before flight.
WATCH: Never use public roads or crowds as training corner markers.

SLIDE 03 Figure-eight coordination

Turn while controlling speed and spacing.
Plan two loops and a central crossing reference.
Maintain visible clearance across the entire pattern.
Slow down before turns instead of correcting too late.
DO: Instructor demonstrates in simulation first.
WATCH: Do not force a figure-eight beyond safe visual range.

MODULE 04
PRESENTATION
Module 04 / Slides 4-6

Orientation and maneuvers
SLIDE 04 Nose-in confidence

Recover orientation without guessing.
Stop translation and stabilize before making a correction.
Use familiar reference points and known heading.
If confused, call HOLD and request assistance early.
DO: Practice 20-second nose-in hover in simulator.
WATCH: Disorientation is not a reason to accelerate.

SLIDE 05 Precision landing

Arrival geometry matters more than speed.
Choose a clear, repeatable approach path.
Reduce drift, align and descend with margins.
Judge accuracy only after safe touchdown.
DO: Compare hard fast arrivals with smooth controlled arrivals.
WATCH: Landing inside the target must never override safety.

SLIDE 06 Flight discipline

Quality is repeatable, not dramatic.
State next movement before executing.
Monitor battery and time against the plan.
Stop and reset after any unstable maneuver.
DO: Coach provides verbal prompts rather than stick intervention when safe.
WATCH: Clean control comes before advanced speed.

MODULE 04
PRACTICAL ACTIVITY
Three-pattern precision circuit
Module 04 | Est. 145 min including rotation

TASK / DEMONSTRATION
Fly defined patterns and manage nose-in orientation with precision.
Equipment and setup
Marked square, target landing pad, simulator and training log.
Procedure
Complete nose-in recovery scenario in simulator.
1
Fly square route with controlled pauses at four corners.
2
Fly a smooth figure-eight within the flight box.
3
Return to center, hold heading and land on target.
4
Repeat once focusing on smoothness and checklist adherence.
5
EVIDENCE TO SUBMIT
Observed precision scorecard and two flight tracks or observer records.
PASS CONDITION
Route remains inside approved flight box; no critical correction; controlled landing within 1 m when safe.
DEBRIEF: Which maneuver most affected your spatial awareness?
SAFETY LIMIT: Use a larger landing tolerance when wind or safety conditions make 1 m unsuitable.

MODULE 04
ASSESSMENT

Module 04 / Knowledge check

Circle one best answer. 5 questions x 1 point. No guessing in flight-critical tasks.
Name _____ Date _____
1 Why is nose-in flight harder for beginners?
A. Battery decreases faster B. Perceived left/right controls may reverse
C. The drone loses GPS D. Motors reverse physically
2 When disoriented, the best first response is:
A. Speed up B. Stabilize/hold and identify orientation
C. Disable sensors D. Turn off the transmitter
3 A precision circuit measures:
A. Only top speed B. Controlled maneuvering and safety
C. Camera zoom level D. Longest radio range
4 Before a turn in a figure-eight, the pilot should:
A. Increase speed sharply B. Manage speed and space
C. Ignore the boundary D. Close eyes to feel controls
5 If a target landing conflicts with safety, you should:
A. Force touchdown on target B. Prioritize a safe landing elsewhere
C. Keep hovering indefinitely D. Land beyond the boundary
Written questions check understanding; the activity performance gate determines whether the trainee may progress. Answer
key and rationale are placed in the instructor section.

MODULE 05
LEARNING MODULE
WHEN THINGS GO WRONG
Emergency response
END-OF-MODULE COMPETENCY
Recognize, stabilize, decide and safely recover from common UAV faults.
5 HOURS SIMULATOR + BRIEF
Performance targets
Respond to simulated low battery and control-link issues.
Distinguish RTH limitations from pilot-controlled recovery.
Execute abort procedures and clear communications.
Presentation sequence
01 02 03
Emergency priorities Low battery Link loss
04 05 06
Position and sensor faults The abort ladder After the incident
ADVANCE ONLY WHEN
Correctly resolve all critical safety decisions; no mission continuation after unsafe state.

MODULE 05
PRESENTATION
Module 05 / Slides 1-3

Emergency response
SLIDE 01 Emergency priorities

People first. Aircraft second. Mission last.
Prevent injury and preserve safe separation.
Identify the actual state: position, battery, link and mode.
Choose the safest available recovery option.
DO: Use a fast verbal scenario: mission vs people vs aircraft.
WATCH: Do not save the mission at the expense of safety.

SLIDE 02 Low battery

Landing reserve is not optional.
Identify first and critical warnings early.
Decide whether home, a safe nearby landing or immediate landing is appropriate.
Account for wind and transit energy.
DO: Rehearse a decision tree from battery trend data.
WATCH: An automated warning may arrive too late for poor planning.

SLIDE 03 Link loss

Configured behavior determines the outcome.
Know exact aircraft and receiver loss-link settings.
Maintain observer watch and avoid panic commands.
Apply manufacturer-specific procedure if link returns.
DO: Simulate lost control link in approved software only.
WATCH: Never deliberately disconnect controls on a live sortie.

MODULE 05
PRESENTATION
Module 05 / Slides 4-6

Emergency response
SLIDE 04 Position and sensor faults

Automation can degrade without notice.
Recognize GNSS or vision-system warnings.
Avoid mode changes that require invalid sensors.
Reduce workload and recover only within demonstrated pilot capability.
DO: Practice sensor degradation in simulator.
WATCH: RTH may not function with unreliable position.

SLIDE 05 The abort ladder

Know your exit before launch.
Cancel tasking, regain stable control, select safe route, land.
Call ABORT / HOLD / LAND with observer confirmation.
Record event after the aircraft is secure.
DO: Rehearse radio-style call-and-response.
WATCH: Do not improvise a new flight profile mid-emergency.

SLIDE 06 After the incident

Learning is part of safety.
Account for all people and any damage.
Secure aircraft and batteries before troubleshooting.
Document event and follow mandatory reporting rules if applicable.
DO: Complete a mock near-miss report in three sentences.
WATCH: Avoid restarting before fault cause is understood.

MODULE 05
PRACTICAL ACTIVITY
Emergency decision simulator circuit
Module 05 | Est. 90 min

TASK / DEMONSTRATION
Recognize, stabilize, decide and safely recover from common UAV faults.
Equipment and setup
Flight simulator, event cards, crew radio/callout sheet.
Procedure
Start from a stable simulated mission with known settings.
1
Run three randomized scenarios: low battery, link loss, navigation fault.
2
For each: announce state; identify options; execute appropriate simulated recovery.
3
Brief the crew and complete a short incident/near-miss record.
4
EVIDENCE TO SUBMIT
Simulator event log, instructor observation card and incident record.
PASS CONDITION
Correctly resolve all critical safety decisions; no mission continuation after unsafe state.
DEBRIEF: What information was essential within the first ten seconds?
SAFETY LIMIT: Never induce loss link, disable GNSS or defeat failsafes on a live training aircraft.

MODULE 05
ASSESSMENT

Module 05 / Knowledge check

Circle one best answer. 5 questions x 1 point. No guessing in flight-critical tasks.
Name _____ Date _____
1 An automatic RTH is safest when:
A. Its prerequisites and path are valid B. Every sensor is failed
C. Home was never recorded D. The pilot has not checked altitude
2 Low battery during a distant mission means:
A. Continue until auto landing B. Reassess reserve and use safest landing plan
C. Increase speed regardless of wind D. Stop monitoring battery
3 A GNSS warning may affect:
A. Only audio recording B. Position-dependent modes and RTH
C. Propeller color D. File names
4 Safe training for lost link is:
A. Switch off transmitter during live flight B. Use an approved simulator scenario
C. Unplug the receiver airborne D. Remove antenna during flight
5 Which priority is correct?
A. Mission, drone, people B. Drone, data, people
C. People, aircraft, mission D. Camera, mission, people
Written questions check understanding; the activity performance gate determines whether the trainee may progress. Answer
key and rationale are placed in the instructor section.

MODULE 06
LEARNING MODULE
CAMERA AND AERIAL DATA
Useful imagery
END-OF-MODULE COMPETENCY
Capture an organized and interpretable aerial documentation set.
4 HOURS FIELD + REVIEW
Performance targets
Choose nadir vs oblique framing for a specific information need.
Control exposure, image sharpness and gimbal orientation.
Label, inspect and protect geotagged mission data.
Presentation sequence
01 02 03
Collect evidence, not scenery Nadir and oblique Avoid image failure
04 05 06
Gimbal and flight path Metadata and chain of custody Quality gate
ADVANCE ONLY WHEN
Images answer the objective, are legible, correctly labeled and ethically collected.

MODULE 06
PRESENTATION
Module 06 / Slides 1-3

Useful imagery
SLIDE 01 Collect evidence, not scenery

Every image should answer a question.
Define subject, required detail and final deliverable.
Plan viewpoint, time of day and glare risk.
Use overlap where continuity or mapping is needed.
DO: Compare a scenic frame against an inspection-ready frame.
WATCH: Attractive footage is not necessarily useful evidence.

SLIDE 02 Nadir and oblique

Two views support different tasks.
Nadir: ground area, roofs and map coverage.
Oblique: facades, damage evidence and feature context.
Capture orientation/scale references when practical.
DO: Match six sample features to best view type.
WATCH: Do not imply oblique stills are survey-grade maps.

SLIDE 03 Avoid image failure

Fix quality at capture time.
Prevent motion blur; hold stable before exposure.
Check focus, exposure and shadow detail.
Avoid aggressive digital zoom for evidence detail.
DO: Repeat a test scene with wrong then corrected exposure.
WATCH: Unrecoverable blur cannot be fixed reliably later.

MODULE 06
PRESENTATION
Module 06 / Slides 4-6

Useful imagery
SLIDE 04 Gimbal and flight path

Keep capture consistent.
Choose safe height and route for subject coverage.
Use repeatable angle and heading for comparisons.
Maintain visual observer and terrain clearance.
DO: Sketch a four-shot capture plan.
WATCH: Camera attention must not replace aircraft observation.

SLIDE 05 Metadata and chain of custody

Good files remain traceable.
Record time, operator, site and mission purpose.
Preserve originals and separate edited copies.
Check geotags; do not assume they are always accurate.
DO: Rename files by mission, date, location and sequence.
WATCH: Do not share images containing private details without authorization.

SLIDE 06 Quality gate

Review before leaving the site.
Confirm complete coverage and no missing areas.
Inspect representative images for sharpness and exposure.
Back up the dataset and note limitations.
DO: Perform a five-image quality review with a partner.
WATCH: Missing evidence is harder to recover after demobilization.

MODULE 06
PRACTICAL ACTIVITY
Five-image evidence set
Module 06 | Est. 70 min

TASK / DEMONSTRATION
Capture an organized and interpretable aerial documentation set.
Equipment and setup
Camera drone or supplied image set, mission brief, review laptop.
Procedure
Choose inspection or site-observation objective.
1
Plan three oblique and two nadir captures (adapt to subject).
2
Acquire images in a permitted site or use instructor dataset.
3
Create a short contact sheet and annotate two findings.
4
Name and back up the source files.
5
EVIDENCE TO SUBMIT
Five-image contact sheet, labeled originals, two useful observations and one limitation.
PASS CONDITION
Images answer the objective, are legible, correctly labeled and ethically collected.
DEBRIEF: Which photo carries the most decision-useful information?
SAFETY LIMIT: Respect privacy, restricted sites and minimum safe separation.

MODULE 06
ASSESSMENT

Module 06 / Knowledge check

Circle one best answer. 5 questions x 1 point. No guessing in flight-critical tasks.
Name _____ Date _____
1 A nadir image is primarily viewed:
A. Toward the horizon B. Straight downward
C. Through thermal only D. From inside the aircraft
2 For repeated facade inspection, prioritize:
A. Random angles B. Consistent oblique framing
C. Maximum digital zoom D. No image labels
3 The best way to handle blurred evidence is:
A. Claim it is sharp B. Retake when safe
C. Delete all metadata D. Sharpen until details appear
4 What should be preserved?
A. Only heavily edited images B. Original source images with context
C. Only thumbnail previews D. Nothing after delivery
5 Before leaving the site, the team should:
A. Assume all shots worked B. Check image quality and coverage
C. Format both memory cards D. Publish raw imagery online
Written questions check understanding; the activity performance gate determines whether the trainee may progress. Answer
key and rationale are placed in the instructor section.

MODULE 07
LEARNING MODULE
AUTONOMOUS FLIGHT AND MAPPING
Waypoints and outputs
END-OF-MODULE COMPETENCY
Plan and supervise a safe waypoint/survey mission and inspect map outputs.
5 HOURS GCS + DATA LAB
Performance targets
Prepare a mission with safe route, altitude, terrain and abort options.
Verify mission upload and aircraft configuration before autonomous flight.
Understand overlap, GCP/RTK and basic orthomosaic limitations.
Presentation sequence
01 02 03
Automation is supervised flight Waypoint logic Survey coverage
04 05 06
Mission verify From photos to map Deliver evidence with limitations
ADVANCE ONLY WHEN
Safe route and failsafe plan; independently verified upload where supported; usable map output or documented processing attempt.

MODULE 07
PRESENTATION
Module 07 / Slides 1-3

Waypoints and outputs
SLIDE 01 Automation is supervised flight

The pilot remains responsible.
Route planning reduces stick workload but adds planning hazards.
Check obstacles, terrain, people, battery and permission.
Define safe pause, abort and return procedure.
DO: Show a route that looks good on screen but intersects a tower.
WATCH: Uploading a route is not the same as verifying it.

SLIDE 02 Waypoint logic

Every point changes aircraft behavior.
Create takeoff, navigation and return/landing sequence.
Set flight mode, altitude frame, speed and expected action.
Check route continuity and safe starting position.
DO: Create a four-point route in QGroundControl.
WATCH: Planned Home may differ from aircraft actual Home.

SLIDE 03 Survey coverage

Image geometry determines map quality.
Draw area polygon and grid.
Specify front/side overlap appropriate to imagery and terrain.
Check camera triggering, mission time and terrain following.
DO: Use QGroundControl Survey pattern on a sample rectangle.
WATCH: Terrain elevation variation can invalidate overlap assumptions.

MODULE 07
PRESENTATION
Module 07 / Slides 4-6

Waypoints and outputs
SLIDE 04 Mission verify

The vehicle must match the plan.
Save versioned plan; upload to correct vehicle.
Read back/confirm mission if supported.
Review altitude, return action, geofence and battery margin.
DO: Have a partner perform a blind mission checklist.
WATCH: Do not launch an unverified or stale plan.

SLIDE 05 From photos to map

Photogrammetry needs quality input.
Inspect blur, geotags and coverage.
Process images to orthomosaic and optional surface/point cloud.
Use GCPs/RTK for accuracy requirements when appropriate.
DO: Review a sample WebODM output and missing-coverage artifact.
WATCH: A pretty map is not proof of survey accuracy.

SLIDE 06 Deliver evidence with limitations

Quality includes honesty.
Export map with coordinate system and date.
Note coverage gaps, georeferencing method and accuracy limits.
Attach a concise mission log and QA checklist.
DO: Inspect map distortion around edges.
WATCH: Do not claim centimeter accuracy without validation.

MODULE 07
PRACTICAL ACTIVITY
Plan - verify - map mini mission
Module 07 | Est. 110 min

TASK / DEMONSTRATION
Plan and supervise a safe waypoint/survey mission and inspect map outputs.
Equipment and setup
QGroundControl or Mission Planner, simulator/aircraft, sample dataset, WebODM/QGIS.
Procedure
Create a fenced 4-waypoint route and identify terrain/obstacle hazards.
1
Create a survey grid over a sample plot and estimate mission time.
2
Save the plan; verify route, altitude frame and intended vehicle.
3
Execute in simulator or supervised authorized flight.
4
Process sample/collected photos to an orthomosaic and mark two quality issues.
5
EVIDENCE TO SUBMIT
Versioned mission file, verification checklist, map screenshot/export and limitations note.
PASS CONDITION
Safe route and failsafe plan; independently verified upload where supported; usable map output or documented
processing attempt.
DEBRIEF: What breaks when planned altitude ignores terrain?
SAFETY LIMIT: Do not fly automated routes until platform-specific checks and trainee gates are passed.

MODULE 07
ASSESSMENT

Module 07 / Knowledge check

Circle one best answer. 5 questions x 1 point. No guessing in flight-critical tasks.
Name _____ Date _____
1 A survey grid is mainly used to:
A. Increase radio power B. Create systematic image coverage
C. Calibrate motors D. Hide obstacles
2 A planned home marker in GCS is:
A. Always the armed home B. A planning reference that must be checked
C. The only safe landing point D. Proof of permission
3 Terrain following is important because:
A. It makes drones silent B. Ground elevation varies below flight path
C. It adds battery capacity D. It removes all obstacles
4 An orthomosaic can be described as:
A. A stitched, georeferenced image product B. A battery log
C. An FPV radio system D. A live collision sensor
5 An accurate map claim requires:
A. A good-looking image B. Appropriate ground checks and documented accuracy
C. Maximum brightness D. No metadata
Written questions check understanding; the activity performance gate determines whether the trainee may progress. Answer
key and rationale are placed in the instructor section.

MODULE 08
LEARNING MODULE
NEXT-GENERATION UAV TECHNOLOGY
AI, RTK, sensors and autonomy
END-OF-MODULE COMPETENCY
Select modern UAV technologies for an application and articulate limitations.
3 HOURS DEMO + DATA LAB
Performance targets
Explain where AI-assisted perception helps and fails.
Distinguish RTK, GNSS, thermal, LiDAR and standard RGB data.
Identify additional safety and authorization issues for remote/autonomous systems.
Presentation sequence
01 02 03
2026 capability map AI in the loop Navigation beyond basic GNSS
04 05 06
Different sensors, different answers Connected autonomous systems Practical technology decision
ADVANCE ONLY WHEN
All selections are mission-relevant, limitations honest, no unsupported autonomy claims.

MODULE 08
PRESENTATION
Module 08 / Slides 1-3

AI, RTK, sensors and autonomy
SLIDE 01 2026 capability map

Buy outcomes, not buzzwords.
AI vision can assist search, tracking and inspection review.
RTK improves positioning when corrections and setup are valid.
Thermal and LiDAR answer different sensing questions.
DO: Choose the right sensor for a roof, field and utility pole.
WATCH: AI confidence is not verified truth.

SLIDE 02 AI in the loop

Human verification remains essential.
Object detectors produce candidate detections, not proof.
Evaluate false positives, false negatives and image conditions.
Keep human authority over flight and safety decisions.
DO: Audit ten sample frames and mark missed detections.
WATCH: No automatic pursuit of people in training exercises.

SLIDE 03 Navigation beyond basic GNSS

Redundancy brings complexity.
GNSS positioning may degrade near structures or canopy.
Visual/inertial aiding can help when fitted and validated.
RTK needs adequate correction, geometry and data workflow.
DO: Compare sample position traces: GNSS vs corrected survey.
WATCH: Do not promise reliable GNSS-denied navigation on every drone.

MODULE 08
PRESENTATION
Module 08 / Slides 4-6

AI, RTK, sensors and autonomy
SLIDE 04 Different sensors, different answers

Know what the dataset actually measures.
RGB captures appearance and texture.
Thermal measures radiance/temperature patterns with caveats.
LiDAR estimates 3D range and geometry.
DO: Match sample data to an inspection question.
WATCH: Thermal imagery is not an x-ray or definitive defect diagnosis.

SLIDE 05 Connected autonomous systems

Remote operations require greater assurance.
Drone docks and remote dispatch change oversight needs.
Multi-drone/relay links add coordination and failure modes.
BVLOS and remote operations may require special permissions.
DO: Map actors in a drone-in-a-box service workflow.
WATCH: Autonomy never cancels airspace, privacy or risk duties.

SLIDE 06 Practical technology decision

Match capability to hazard, cost and evidence.
State mission need, sensor, software and validation method.
Show a low-cost fallback and failure case.
Separate proven workflows from experimental demonstrations.
DO: Teams pitch one upgrade in 60 seconds.
WATCH: Reject innovation that lacks an identifiable use case.

MODULE 08
PRACTICAL ACTIVITY
Technology-selection challenge
Module 08 | Est. 45 min

TASK / DEMONSTRATION
Select modern UAV technologies for an application and articulate limitations.
Equipment and setup
Sample RGB, thermal and LiDAR outputs; AI detection examples; decision matrix.
Procedure
Receive three missions: missing person mock search, roof inspection and terrain mapping.
1
Select aircraft/payload/navigation/data-processing features for each.
2
Identify two limitations and one human verification step per mission.
3
Present an affordable baseline vs premium configuration.
4
EVIDENCE TO SUBMIT
Completed technology-selection matrix and brief rationale.
PASS CONDITION
All selections are mission-relevant, limitations honest, no unsupported autonomy claims.
DEBRIEF: Which technology offers the most value per peso for your mission?
SAFETY LIMIT: Special permissions and risk management apply to non-standard or remote operations.

MODULE 08
ASSESSMENT

Module 08 / Knowledge check

Circle one best answer. 5 questions x 1 point. No guessing in flight-critical tasks.
Name _____ Date _____
1 AI object detection output should be treated as:
A. A verified fact B. A hypothesis requiring checking
C. Proof of legal permission D. A substitute for a spotter
2 RTK primarily helps improve:
A. Positioning under valid corrections B. Motor temperature
C. Image compression D. Camera aperture
3 Thermal imagery is most suitable for:
A. Measuring spectral appearance/heat patterns B. Recording sound
C. Measuring exact ground distance directly D. Controlling propeller pitch
4 LiDAR principally provides:
A. Radio frequencies B. 3D distance/geometry information
C. Battery charging D. Audio transcription
5 A remote drone dock automatically makes BVLOS legal:
A. Always B. Only for small drones
C. No, operational authorization still applies D. Only after sunset
Written questions check understanding; the activity performance gate determines whether the trainee may progress. Answer
key and rationale are placed in the instructor section.

MODULE 09
LEARNING MODULE
THE REAL-WORLD MISSION
Plan, fly, deliver
END-OF-MODULE COMPETENCY
Complete a supervised mission from tasking through evidence-based reporting.
4 HOURS TEAM FIELD MISSION
Performance targets
Use a briefing, risk card and aircraft checklist as a team.
Collect useful data within a defined mission boundary.
Deliver findings, data files and a documented limitation statement.
Presentation sequence
01 02 03
Mission brief to purpose Crew roles Mission card
04 05 06
Fly and monitor Verify the data Debrief like an operator
ADVANCE ONLY WHEN
Complete record, no critical hazard, two useful findings and transparent limitations.

MODULE 09
PRESENTATION
Module 09 / Slides 1-3

Plan, fly, deliver
SLIDE 01 Mission brief to purpose

Translate an ask into a deliverable.
Choose site survey, mock damage assessment, field observation or inspection.
Define beneficiary, outputs, coverage and acceptance test.
Set timeline, operating limits and risk constraints.
DO: Turn a vague request into a two-line mission statement.
WATCH: Do not fly before the information need is clear.

SLIDE 02 Crew roles

One mission, clear authority.
Assign pilot, observer, data lead and safety lead (as available).
Use closed-loop communications on hazards and commands.
Clarify abort authority and changes in control.
DO: Run a 90-second role brief.
WATCH: No overlapping or conflicting pilot control.

SLIDE 03 Mission card

Write the plan before the flight.
Site, mission, flight box, times and weather.
Aircraft, battery, planned route and contingencies.
Evidence list and privacy/data-handling plan.
DO: Fill one-page mission card and conduct peer challenge.
WATCH: Unreviewed changes require a new GO / NO-GO.

MODULE 09
PRESENTATION
Module 09 / Slides 4-6

Plan, fly, deliver
SLIDE 04 Fly and monitor

Observe the mission, not just the camera.
Announce takeoff, checkpoints, warnings and landing.
Monitor position, margins, battery, people and airspace.
Abort if a critical assumption fails.
DO: Instructor injects harmless verbal scenario change.
WATCH: Do not force completion when conditions change.

SLIDE 05 Verify the data

A mission is unfinished without usable output.
Check images/logs while still at site.
Identify gaps, re-fly only if safe and authorized.
Save originals and record data limitations.
DO: Use a short QA checklist before demobilizing.
WATCH: Do not invent observations from ambiguous imagery.

SLIDE 06 Debrief like an operator

Close the loop.
Compare planned objectives with delivered outputs.
Record time, deviations, anomalies and lessons.
Issue an actionable one-page mission summary.
DO: Each crew member gives one improvement.
WATCH: A clean report includes what you do not know.

MODULE 09
PRACTICAL ACTIVITY
Team mission: inspect, survey or assess
Module 09 | Est. 135 min

TASK / DEMONSTRATION
Complete a supervised mission from tasking through evidence-based reporting.
Equipment and setup
Aircraft or simulator, field site, mission card, image dataset, logs, laptop.
Procedure
Receive a client-like brief and choose a permitted mission scenario.
1
Allocate crew, complete GO / NO-GO and flight briefing.
2
Execute supervised route or capture plan; monitor and call deviations.
3
Perform field data QA and create a one-page mission report.
4
Present two findings and one uncertainty during debrief.
5
EVIDENCE TO SUBMIT
Mission card, flight log, evidence folder, QA sheet and final 1-page report.
PASS CONDITION
Complete record, no critical hazard, two useful findings and transparent limitations.
DEBRIEF: What decision could the client make using your output?
SAFETY LIMIT: Use mock disaster sites and obtain land/site permission; avoid actual hazardous response deployment.

MODULE 09
ASSESSMENT

Module 09 / Knowledge check

Circle one best answer. 5 questions x 1 point. No guessing in flight-critical tasks.
Name _____ Date _____
1 A good mission brief begins with:
A. Desired deliverable and decision B. Maximum drone speed
C. A logo D. Video music
2 A safety observer primarily helps:
A. Choose music B. Monitor hazards, traffic and aircraft
C. Rewrite the map projection D. Charge a flying battery
3 A newly identified critical hazard requires:
A. Continue unchanged B. Reassess and potentially abort
C. Ignore if images are good D. Fly higher immediately
4 A final mission report must include:
A. Only successes B. Evidence, limitations and deviations
C. Only the drone serial number D. Only a map screenshot
5 Good data handling means:
A. Publicly share all imagery B. Maintain organized original files and permissions
C. Delete flight logs D. Rename randomly
Written questions check understanding; the activity performance gate determines whether the trainee may progress. Answer
key and rationale are placed in the instructor section.

MODULE 10
LEARNING MODULE
FINAL FLIGHT CHALLENGE
Demonstrate competency
END-OF-MODULE COMPETENCY
Demonstrate entry-level competence with no critical safety error.
2 HOURS INDEPENDENT CHECKRIDE
Performance targets
Brief and prepare an aircraft without instructor prompting.
Fly a controlled practical sequence and respond to simulated faults.
Produce a complete log and debrief against the rubric.
Presentation sequence
01 02 03
The checkride standard Gate 1: safety Gate 2: control
04 05 06
Gate 3: judgment Gate 4: autonomy and data Close the loop
ADVANCE ONLY WHEN
At least 80% overall and PASS every safety-critical item; five hours of supervised live flight logged separately
as a program target.

MODULE 10
PRESENTATION
Module 10 / Slides 1-3

Demonstrate competency
SLIDE 01 The checkride standard

Reliability matters more than showmanship.
Know the mission, hazards, aircraft and pilot limits.
Prioritize repeatability, safe margins and communication.
Any unsafe condition means pause or abort.
DO: Review assessed competencies before trainee begins.
WATCH: The examiner never rewards unnecessary risk.

SLIDE 02 Gate 1: safety

Verify a real GO / NO-GO decision.
Site and airspace review; weather and crew roles.
Preflight, battery, RTH and modes.
Correct reject/abort when a critical hazard exists.
DO: Use one hidden defect card.
WATCH: Critical misses override numeric scores.

SLIDE 03 Gate 2: control

Demonstrate essential flight skill.
Safe takeoff, hover and four-direction translation.
Controlled turn/precision element at trainee level.
Safe return and landing.
DO: Assessor observes without unnecessary prompting.
WATCH: Flight difficulty is adjusted for safe conditions.

MODULE 10
PRESENTATION
Module 10 / Slides 4-6

Demonstrate competency
SLIDE 04 Gate 3: judgment

Think while under controlled pressure.
Respond verbally to two emergency scenarios.
Choose a safe recovery or abort method.
State when RTH cannot be trusted.
DO: Use simulator or oral decision card after landing.
WATCH: Never introduce faults in actual flight.

SLIDE 05 Gate 4: autonomy and data

Show a usable mission workflow.
Inspect a waypoint plan and verify correct vehicle.
Identify one route hazard and one QA concern.
Deliver map/image example with limitations.
DO: Short desktop GCS verification plus output review.
WATCH: Tool familiarity does not imply unrestricted autonomy.

SLIDE 06 Close the loop

Assessment is evidence of current skill.

Log the observed competency for each trainee.
Reassess failed gates after targeted practice.
Issue institutional completion only when requirements met.
DO: Conduct individual strengths and gaps debrief.
WATCH: Course completion is not a regulator-issued certificate.

MODULE 10
PRACTICAL ACTIVITY
Individual competency checkride
Module 10 | Est. 100 min scheduled rotation

TASK / DEMONSTRATION
Demonstrate entry-level competence with no critical safety error.
Equipment and setup
Approved drone, simulator, scenario cards, checkride rubric and flight log.
Procedure
Pass safety/aircraft preflight gate.
1
Perform takeoff, hover, precision maneuver and safe landing.
2
Respond to two simulator or oral emergency events.
3
Verify one waypoint plan and inspect one sample map/data output.
4
Complete the log, receive debrief and sign record.
5
EVIDENCE TO SUBMIT
Completed signed checkride sheet, simulator log and final flight record.
PASS CONDITION
At least 80% overall and PASS every safety-critical item; five hours of supervised live flight logged separately as a
program target.
DEBRIEF: What is the next flight skill you should train before solo work?
SAFETY LIMIT: A trainee who fails a critical item must be stopped, coached and reassessed; no automatic pass.

MODULE 10
ASSESSMENT

Module 10 / Knowledge check

Circle one best answer. 5 questions x 1 point. No guessing in flight-critical tasks.
Name _____ Date _____
1 The instructor should prioritize:
A. Speed of completion B. Documented safe competence
C. The most dramatic flight D. Shortest flight log
2 A critical safety item is failed. The result is:
A. Automatic pass above 80% B. Remediation and reassessment
C. Extra credit for speed D. Ignore the item
3 An emergency fault in assessment should be:
A. Induced live without warning B. Simulated or orally assessed
C. Created by disconnecting antennas D. Ignored entirely
4 A flight log should separate:
A. Morning and afternoon only B. Actual aircraft time from simulator time
C. Different pen colors D. Camera resolutions
5 Completion of this course grants:
A. Automatic commercial approval B. An institutional completion record, not CAAP qualification
C. All BVLOS permissions D. Automatic drone registration
Written questions check understanding; the activity performance gate determines whether the trainee may progress. Answer
key and rationale are placed in the instructor section.

INSTRUCTOR GUIDE
Delivery rules and constraints
Use this as a training package - validate against local conditions.
BEFORE FIRST COHORT
Verify applicable CAAP rules, provider approvals, operating sites, aircraft manuals, insurance and internal training SOPs.
MC 064-2026 is relevant to RPAS ground/flight training by ROC/COA holders; obtain legal/regulatory confirmation
before advertising a regulated qualification.
RATIO AND ROTATIONS
Recommend one trainer-managed aircraft per two trainees when practical, with visual observer coverage. Ru three
parallel stations (simulator / aircraft / data lab) to minimize idle time and allow individual logs.
ASSESSMENT INTEGRITY

A 5-item multiple-choice check is formative, not sufficient for flight approval. Keep critical competency gates as observed
PASS / RETRAIN. Document attempts, remediation and retests.
EMERGENCY PRACTICE
Never cause a live lost-link, GNSS failure or battery-critical condition. Simulate faults or use controlled table-top prompts.
Test aircraft failsafe configuration only per approved manufacturer/firmware procedures.
HARDWARE DIFFERENCES
Controller layouts, aircraft modes, RTH behavior and battery thresholds vary by model. Replace generic controls or
recommended training dimensions with model-specific instructions and local risk limits.
DATA AND PRIVACY
Collect minimum required data, control access to imagery, preserve originals, and document metadata and uncertainty.
Avoid sensitive areas or identifiable people without consent/authority.

FIELD RESOURCE
Before launch / crew checklist
Use with the actual approved aircraft manual and location.
AUTHORIZATION AND SITE
Operation category, permits and required crew qualifications verified.
Airspace, temporary restrictions and local approval checked.
Weather / wind / visibility within aircraft and instructor limits.
Bystanders controlled; safe flight box and emergency landing options ready.
AIRCRAFT AND POWER
Airframe, arms, props, payload and battery physically sound.
Controller link, correct flight mode and aircraft status reviewed.
Home point, RTH height, geofence and applicable failsafe state verified.
Battery reserve and planned mission duration meet policy.
CREW AND FINAL CLEARANCE
Pilot, observer and data roles clearly assigned.
Brief commands, lost-link and abort actions confirmed.
Takeoff/landing zone physically clear; spotter confirms CLEAR.
GO / NO-GO decision announced and recorded.
Pilot / trainee _____ Observer _____ Date _____

FIELD RESOURCE
After landing / data checklist
Finish the operation before packing the drone.
AIRCRAFT SECURE
Touchdown confirmed; motors disarmed; propellers stopped.
Power isolated per aircraft procedure; battery condition checked.
Aircraft inspected for damage or unusual vibration/heat.
Any warnings, defects, near-misses and maintenance needs recorded.
EVIDENCE AND DELIVERABLES
Mission objective and collection coverage checked.
Original imagery, log and metadata saved to approved storage.
Image quality checked for blur, exposure and omissions.
Data access, privacy and storage policy applied.
CREW CLOSURE
Flight times separated into live and simulator hours.
Site cleared of equipment and battery hazards.
Debrief includes improvement, deviation and follow-up owner.
Supervisor signs completed flight record.
Pilot / trainee _____ Observer _____ Date _____

EVALUATION FORM
Individual competency checkride
Mark every critical task PASS / RETRAIN before awarding a numeric score.
Trainee: _____ Aircraft: _____
COMPETENCY WEIGHT SCORE
Site/airspace GO / NO-GO and crew brief
25% ____ / ____
Airframe, battery and failsafe inspection
Gate ____ / ____
Takeoff, hover, translation, precision landing
30% ____ / ____
Simulated emergency response
20% ____ / ____
Mission plan verification and data QA
25% ____ / ____
CRITICAL GATES - PASS / RETRAIN
Airspace and site legality reviewed P/R
No active critical aircraft warning / defect ignored P/R
Safe clearance and positive preflight call P/R
No unsafe pilot action or instructor intervention P/R
Safe landing or immediate abort when required P/R
FINAL: TOTAL _____ / 100 CRITICAL GATES _____ PASS / RETRAIN
Assessor signature _____ Date _____

EVALUATION FORM
Flight exposure and incident log
Individual actual-aircraft and simulator time must be recorded separately.
Trainee: _____ Cohort: _____
DATE AIRCRAFT SKILL / MISSION LIVE MIN SIM MIN SUPERVISOR
ACTUAL-AIRCRAFT TOTAL _____ min SIMULATOR TOTAL _____ min
ANOMALY / INCIDENT RECORD
Event time / place
Observed state / action
People / property effects
Immediate response / follow-up
Reviewed by _____ Date _____

SOURCE NOTES
Official references and verification
Selected authoritative sources; checked 30 October 2026.
CAAP / RPAS Regulations
Regulatory framework: PCAR Parts 1, 2, 4, 11
OPEN OFFICIAL RESOURCE
CAAP / Types of Operations
Commercial/non-commercial and special flight permit pathways
OPEN OFFICIAL RESOURCE
CAAP / MC 064-2026
Guidelines for RPAS ground and flight training by ROC/COA holders
OPEN OFFICIAL RESOURCE
ArduPilot / Pre-flight Checklist
Flight readiness and gradual flight mode introduction
OPEN OFFICIAL RESOURCE
ArduPilot / Radio & Battery Failsafes
Configurable link-loss and low-battery behavior
OPEN OFFICIAL RESOURCE
ArduPilot / Battery Failsafe
Power monitoring and safe fallback actions
OPEN OFFICIAL RESOURCE
QGroundControl / Plan View
Mission planning, upload and waypoint review
OPEN OFFICIAL RESOURCE
QGroundControl / Survey Pattern
Grid survey planning and terrain-following cautions
OPEN OFFICIAL RESOURCE
OpenDroneMap / Outputs
Orthophoto, point cloud and digital elevation products
OPEN OFFICIAL RESOURCE
EASA / SORA 2.5
International risk assessment learning reference, not Philippine law
OPEN OFFICIAL RESOURCE